

# Reachability in Linear Recurrence Automata

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**Abstract.** This paper studies the linear recurrence automata model in which multiple linear recurrences can be used to generate sequences of numbers. We parameterised the model by varying several structural properties to identify under which constraints or extensions the reachability problems become computationally hard and undecidable. We show first the reduction of classical matrix semigroup problems to reachability in linear recurrence automata and then analyse several variants of the model restricting the state structures, the number and the depth of recurrences.

**Keywords:** Linear recurrences · Matrix Semigroups · Reachability

## 1 Introduction

Reachability problems in linear recurrence sequences represent a fundamental area of research with wide-ranging applications in theoretical computer science, mathematics, population dynamics, etc [3,5,6,11]. Reachability in linear recurrence sequences is related to Diophantine approximation of the irrational numbers, matrix equations, regular grammars and finite state machines. Although significant progress has been made in understanding the dynamics of linear recurrence sequences, there exist many challenges and open problems in this area [4,13,15]. This paper studies nondeterministic automata model when more than one linear recurrence can be used to generate sequences and we aim to understand under which constraints the reachability problems for such systems become computationally hard, undecidable or equivalent to other problems.

One notable example of reachability problems in linear recurrence sequences is the renowned Skolem-Pisot problem [5]. This is the problem of determining whether a given linear recurrent sequence includes the number zero. In the biological context the linear recurrence describes the population dynamics of species when the age-specific survival rate is equal to 1. In more general form, the sub-diagonal of 1's in the companion matrix of a linear recurrence can be replaced by the age-specific survival rates  $s_1, s_2, \dots$  then such matrix  $M$  is known in biological literature as a Leslie Matrix [6]. For example the matrix product  $M_1^p M_2^q$  may correspond to a case where a population which is governed by two different sets of fertility and survival parameters represented by  $M_1$  for one “season” within  $p$

discrete time steps and  $M_2$ , e.g. for one in the next “season” (or a change in environmental conditions) for  $q$  steps [7]. It is natural to consider a more complex dynamics where the order of changes in biological or technological processes can be described by automata structure and formulated as reachability questions.

The reachability problems with several linear recurrences has not been studied in the literature and in this paper we introduce and study the model of automata with linear recurrence updates and generalised reachability subsequence problem<sup>4</sup>. The simplest form of such automata is an automata with one state and one linear recurrence. The reachability of subsequence of length 1 which is equal to zero with single linear recurrence corresponds to the Skolem-Pisot problem which decidability status in general is still open. The generalisation of linear recurrence system can be achieved by increasing the number, depth and order of applications of linear recurrences. In this paper we show that in the contrast to stateless systems like Skolem problem, enriching the structure with states and additional recurrences leads to undecidability of the reachability problem. In Section 2 the reductions from several matrix semigroup problems can be used to identify under which parameters the reachability problems become undecidable. In Section 3 is focused on the limitations to the state structure where it was shown that the single loop system with multiple linear recurrences is equivalent to the system with a single linear recurrence and the sequence linearly connected loops still lead to undecidability. In Section 4 provides motivation to study non-deterministic recurrence systems of the order two, by showing the PSPACE hardness in this case as well as decidability for the order one systems and undecidability for the order three. Finally in Section 5 we provide a new model of linear recurrences over matrices generalising the original concept.

## 2 Linear recurrence sequences and automata

Possibly the simplest, and certainly the most well known linear recurrence sequence is the *Fibonacci sequence*. It is the sequence  $(x_t)_{t \in \mathbb{N}} \in \mathbb{Z}^{\mathbb{N}}$  defined by the initial conditions  $x_1 = 1$  and  $x_0 = 0$  and the linear recurrence  $x_t = x_{t-1} + x_{t-2}$ , or equivalently,

$$\begin{pmatrix} x_t \\ x_{t-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} x_{t-1} \\ x_{t-2} \end{pmatrix}, \quad (1)$$

for  $t \in \mathbb{N} \setminus \{0, 1\}$ . In general, a *linear recurrence* of *depth* (or *order*)  $k \in \mathbb{N}$  over a field  $\mathbb{F}$  is given by a  $k$ -tuple  $a = (a_1, \dots, a_k) \in \mathbb{F}^k$ , which we regard as a recursive definition

$$x_t = a_1 \cdot x_{t-1} + \dots + a_k \cdot x_{t-k}, \quad (2)$$

for  $t \in \mathbb{N} \setminus \{0, \dots, k-1\}$ , of a sequence  $(x_t)_{t \in \mathbb{N}} \in \mathbb{F}^{\mathbb{N}}$  starting with an initial condition  $(x_{k-1}, \dots, x_0)$ . This recurrence (1) can be written as

$$\begin{pmatrix} x_t \\ \vdots \\ x_{t-k+1} \end{pmatrix} = C(a) \cdot \begin{pmatrix} x_{t-1} \\ \vdots \\ x_{t-k} \end{pmatrix} \quad (3)$$

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<sup>4</sup> The similar model name with a focus on a language recognition was discussed in [10]

using the *companion matrix* of  $a$ :

$$C(a) = \begin{pmatrix} a_1 & a_2 & \cdots & a_{k-1} & a_k \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & & 0 & 0 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}. \quad (4)$$

We may generalize this further and consider (nondeterministic) automata with transitions labelled by linear recurrences. A *linear recurrence automaton*  $\mathcal{A} = (G, a, q_0, Q_{\text{acc}})$  of *depth*  $k \in \mathbb{N}$  over a field  $\mathbb{F}$  consists of

- a directed graph  $G = (Q, E, \delta^-, \delta^+)$  (possibly with self-loops and parallel edges), i.e., finite sets  $Q$  and  $E$  whose elements are called *states* and *transitions*, respectively, together with a pair of functions  $\delta^-, \delta^+ : E \rightarrow Q$  specifying that a transition  $e \in E$  is from state  $\delta^-(e)$  to state  $\delta^+(e)$ ;
- for each edge  $e \in E$ , a linear recurrence  $a^{(e)} \in \mathbb{F}^k$  of depth  $k$ ;
- a vertex  $q_0 \in Q$  called the *initial state*; and
- a set  $Q_{\text{acc}} \subseteq Q$  of *final states*.

A *walk* in this graph  $G$  is a sequence  $p = (e_0, e_1, \dots, e_{|p|-1}) \in E^*$  of transitions such that  $\delta^-(e_t) = \delta^+(e_{t-1})$  for each  $t = 1, \dots, |p| - 1$ . We say that  $p$  is *from* the state  $\delta^-(e_0)$  and *to* the state  $\delta^+(e_{|p|-1})$ . From an initial  $k$ -tuple  $u \in \mathbb{F}^k$  (called an *initial condition*), a walk  $p = (e_k, e_{k+1}, \dots, e_{k+|p|-1}) \in E^{|p|}$  *generates* a unique sequence  $(x_0, \dots, x_{k+|p|-1}) \in \mathbb{F}^{k+|p|}$  such that

- $(x_0, \dots, x_{k-1}) = u$ , and
- for each  $t = k, \dots, |p| - 1$ , we have (3) with  $a = a^{(e_t)}$ .

Repeated application of (3) shows that  $x_t$ , for each  $t \geq k$ , is the first component of the vector

$$C(a^{(e_t)}) \cdot C(a^{(e_{t-1})}) \cdots C(a^{(e_k)}) \cdot \begin{pmatrix} x_{k-1} \\ \vdots \\ x_0 \end{pmatrix}. \quad (5)$$

The automaton  $\mathcal{A}$  is said to *accept* a numerical sequence  $x \in \mathbb{F}^*$  of length  $\geq k$  if  $\mathcal{A}$  has a walk  $p$  from the initial state  $q_0$  to a final state  $\in Q_{\text{acc}}$  such that  $x$  is generated by  $p$  from the initial segment  $x_{<k} = (x_0, \dots, x_{k-1})$  of  $x$ .

**Definition 1.** Let  $s, m, k \in \mathbb{N}$ . We write  $\text{LRA}_{\mathbb{F}}(s, m, k)$  for the class of linear recurrence automata with less than or equal to  $s$  states and less than or equal to  $m$  transitions labeled by linear recurrences of depth  $k$  over  $\mathbb{F}$ .

**Definition 2.** Let  $\mathcal{C}$  be the class of linear recurrence automata of depth  $k \in \mathbb{N}$  over a field  $\mathbb{F}$ . For  $l \in \mathbb{N}$  with  $l \leq k$ , the  $l$ -subsequence reachability problem ( $l$ -SR) for  $\mathcal{C}$  is to decide, for a given  $\mathcal{A} \in \mathcal{C}$ , an initial sequence  $u \in \mathbb{F}^k$  and a

target sequence  $v \in \mathbb{F}^l$ , whether  $\mathcal{A}$  accepts a sequence over  $\mathbb{F}$  starting with  $u$  and ending with  $v$ .<sup>5</sup>

If every state of an automaton  $\mathcal{A} \in \mathcal{C}$  is final, SR corresponds to the problem whether  $v$  can appear in the infinite orbit generated by  $\mathcal{A}$ . Note that if  $\mathcal{A} \in \text{LRA}_{\mathbb{Z}}(1, 1, k)$  (i.e.  $\mathcal{A}$  has a single state and a single linear recurrence of order  $k$ ) and  $v$  is the length-1 sequence  $(0)$ , then the problem corresponds the well-known Skolem-Pisot problem.

**Definition 3.** Let  $\mathbb{F}$  be a field and let  $r, n \in \mathbb{N}$ . Let us denote the semigroup generated by a set  $\mathcal{M} \subseteq \mathbb{F}^{n \times n}$  of matrices over a field  $\mathbb{F}$  by  $\langle \mathcal{M} \rangle$ .

- The  $(r, n)$ -scalar reachability problem over  $\mathbb{F}$  is to decide, given two vectors  $u, v \in \mathbb{F}^n$ , a scalar  $\lambda \in \mathbb{F}$  and a finite set  $\mathcal{M} \subseteq \mathbb{F}^{n \times n}$  of  $r$  matrices, whether there exists  $M \in \langle \mathcal{M} \rangle$  such that  $u^\top \cdot M \cdot v = \lambda$ .
- The  $(r, n)$ -vector reachability problem over  $\mathbb{F}$  is to decide, given two vectors  $u, v \in \mathbb{F}^n$  and a finite set  $\mathcal{M} \subseteq \mathbb{F}^{n \times n}$  of  $r$  matrices, whether there exists  $M \in \langle \mathcal{M} \rangle$  such that  $M \cdot v = u$ .
- The  $(r, n)$ -membership problem over  $\mathbb{F}$  is to decide, given a finite set  $\mathcal{M} \subseteq \mathbb{F}^{n \times n}$  of  $r$  matrices and another matrix  $M \in \mathbb{F}^{n \times n}$ , whether  $M \in \langle \mathcal{M} \rangle$ .

The case with iteration of one linear recurrence (Skolem-Pisot problem) can be represented as the scalar reachability problem for the matrix semigroup generated by a single companion matrix. Here we show the other direction that various matrix reachability problems can be reduced to subsequence reachability problems for linear recurrence automata. By applying Theorem 1 in combination with known hardness or undecidability results on matrix problems can provide a landscape of boundaries for hard cases of linear recurrence automata model. In order to tighten the results more sophisticated constructions could be used like shown in the Section 4.

**Theorem 1.** Let  $\mathbb{F}$  be a field and  $r, n \in \mathbb{N}$ .

- The  $(r, n)$ -scalar reachability problem over  $\mathbb{F}$  can be reduced to 1-SR for

$$\text{LRA}_{\mathbb{F}}(r(n-1) + 2, rn + 1, 2n - 1).$$

- The  $(r, n)$ -vector reachability problem over  $\mathbb{F}$  can be reduced to  $n$ -SR for

$$\text{LRA}_{\mathbb{F}}(r(n-1) + 1, rn, 2n - 1).$$

- The  $(r, n)$ -membership problem over  $\mathbb{F}$  can be reduced to  $n^2$ -SR for

$$\text{LRA}_{\mathbb{F}}(r(n^2 - 1) + 1, rn^2, 2n^2 - 1).$$

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<sup>5</sup> In contrast to standard subsequence reachability on at least one run, it is possible to define the *strong subsequence reachability* problem of deciding whether a particular subsequence can be reached on all possible non-deterministic runs on  $\mathcal{A} \in \mathcal{C}$ .

*Proof.* Let us consider a reduction from the  $(r, n)$ -scalar reachability problem over  $\mathbb{F}$ . Suppose we are given two vectors  $u, v \in \mathbb{F}^n$ , a scalar  $\lambda \in \mathbb{F}$  and a set  $\mathcal{M} \subseteq \mathbb{F}^{n \times n}$  of  $r$  matrices. The problem is to decide whether there exists  $M \in \langle \mathcal{M} \rangle$  such that  $u^\top \cdot M \cdot v = \lambda$ . We will build an automaton in  $\text{LRA}_{\mathbb{F}}(r(n-1)+2, rn+1, 2n-1)$  by decomposing an application of each linear transformation into a sequence of linear recurrences of depth upto  $2n-1$ .

In the product  $u^\top \cdot M \cdot v = \lambda$  let us consider  $M \cdot v$  part. The initial values of linear recurrence will be  $v = (v_1, \dots, v_n) \in \mathbb{F}^n$ . We start the reduction from a set of generators that are not in the form of the companion matrices. So for each  $n \times n$  matrix

$$M = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix} \in \mathcal{M},$$

we consider a directed graph that consists of a path with  $n+1$  states  $q_{M,0}, q_{M,1}, \dots, q_{M,n}$  and, for each  $i \in \{1, \dots, n\}$ , a transition from  $q_{M,i-1}$  to  $q_{M,i}$ , associated the linear recurrences of depth  $n+i-1$  and a vector of coefficients

$$(\underbrace{0, \dots, 0}_{i-1}, a_{in}, \dots, a_{i1}), \text{ corresponding to } x_t = a_{in} \cdot x_{t-i} + \dots + a_{i1} \cdot x_{t-i-n+1}$$

to compute the next  $n$  values in the sequence  $v'_1, \dots, v'_n$  s.t.  $v' = (v'_1, \dots, v'_n)^T = M \cdot v$  with the walk from  $q_{M,0}$  to  $q_{M,n}$  (of length  $n$ ). Now we merge the  $2 \cdot r$  states  $q_{M,0}$  and  $q_{M,n}$ , for all  $M \in \mathcal{M}$ , into a single common initial state  $q_0$ , obtaining a petal graph with  $r$  cycles (which has altogether  $r \cdot (n-1) + 1$  states and  $r \cdot n$  transitions).

Finally for scalar reachability reduction we add to this petal graph one transition from  $q_0$  to a unique final state  $q_f$ , with the linear recurrence corresponding to  $u^\top$ , to compute the value  $\lambda = u^\top \cdot M \cdot v$ . The number of different linear recurrence functions used in the automata is bounded by the number of transitions which is  $rn+1$  and the number of states in the automaton is  $r(n-1)+2$ . If the sequence contains a value  $\lambda$  that is generated by the automaton in the state  $q_f$  then  $\lambda$  can be reached in the scalar reachability problem.

The reduction from the vector reachability (whether there exists  $M \in \langle \mathcal{M} \rangle$  such that  $M \cdot v = u$ ) is identical, with the only difference that, instead of adding this new final state, we simply make  $q_0$  the final state reducing to the question whether  $n$ -subsequence corresponding to the values of the vector  $u$ . Finally for the membership problem, let us denote by  $X$  a matrix that can be generated in the matrix semigroup. The linear recurrence automaton should start with an initial sequence of length  $n^2$  that correspond to values of the matrix  $X$

$$x_{1,1}, \dots, x_{n,1}, x_{1,2}, \dots, x_{n,2}, \dots, x_{1,j}, \dots, x_{n,j}, \dots, x_{1,n}, \dots, x_{n,n}$$

and  $X$  will be equal to identity matrix at the initialisation step. Then each loop of an automata will be a path of  $n^2$  linear recurrences to compute sequentially the next  $n^2$  values in the sequence corresponding to multiplication from the left

$M \cdot X$ . As in the previous construction the initial and final states of the paths corresponding to each matrix in a generator need to be merged to get the petal graph. Then the membership problem for a matrix semigroup corresponds to the  $n^2$ -subsequence reachability problem in constructed automata with  $r(n^2 - 1) + 1$  states,  $rn^2$  transitions with linear recurrences of depth  $2n^2 - 1$ .  $\square$

Theorem 1 can be used to derive undecidability for different classes  $\text{LRA}_{\mathbb{F}}(s, m, k)$  utilizing matrix semigroup results. For example, since the  $(5, 3)$ -scalar reachability problem (i.e., the one for semigroups generated by five integer matrices of dimension 3) is undecidable [9], it follows that 1-SR is undecidable for  $\text{LRA}_{\mathbb{Z}}(12, 16, 5)$ , i.e., for automata having 12 states, 16 distinct linear recurrences of order 5.

**Theorem 2.** *The  $k$ -subsequence reachability problem ( $k$ -SR) for  $\text{LRA}_{\mathbb{Z}}(s, m, k)$  with a single final state can be reduced to the  $(m, k+2)$ -vector reachability problem over  $\mathbb{Z}$ .*

*Proof.* Let  $\mathcal{A} = (G, a, q_0, Q_{\text{acc}}) \in \text{LRA}_{\mathbb{Z}}(s, m, k)$ , where we may assume that the set of states in  $G = (Q, E, \delta^-, \delta^+)$  is  $Q = \{0, \dots, s-1\}$ . For every transition  $e \in E$ , we build a matrix

$$M_e = \begin{pmatrix} C(a^{(e)}) & \bar{0} \\ \bar{0}^T & D_e \end{pmatrix}, \quad (6)$$

where  $C(a^{(e)})$  is the companion matrix of  $a^{(e)}$  (see (4)) and

$$D_e = \begin{pmatrix} s & \delta^+(e) - s \cdot \delta^-(e) \\ 0 & 1 \end{pmatrix}. \quad (7)$$

Then the reachability of  $v \in \mathbb{Z}^k$  from  $u \in \mathbb{Z}^k$  in a state  $q \in Q_{\text{acc}}$  would correspond to the vector reachability of a vector  $(v, q, 1)^T$  from  $(u, q_0, 1)^T$  for  $\langle \{M_e : e \in E\} \rangle$ . To see this, let  $(Q, E)$  be a directed graph with vertices  $q_0, \dots, q_{s-1}$ . Then there exists a set of affine functions  $F$  such that for an edge

$$e = (\delta^-(e), \delta^+(e)) \in E,$$

- there exists a unique  $f_{\delta^-(e)\delta^+(e)} \in F$  such that  $f_{\delta^-(e)\delta^+(e)}(\delta^-(e)) = \delta^+(e)$
- $f_{\delta^-(e)\delta^+(e)}(x) \notin [0, s-1]$  for all  $x \neq \delta^-(e)$  and
- $f_{\delta^-(e)\delta^+(e)}(x) = s \cdot x + b = \delta^+(e)$ , where  $b = \delta^+(e) - s \cdot \delta^-(e) \in [-s^2 + s, s-1]$ .

The idea of the encoding is to represent each state by integers in  $[0, s-1]$  and every edge by affine functions of the form  $ax + b$ , where a value of  $x$  changes according to the automata structure taking as an input the value of one state and computing the value of the resulting state. In the event of application of affine function which is not representing outgoing edge from the current state the coefficient  $s$  ensures that the resulting value of the register will either exceed  $s-1$  or fall below zero. Moreover applying a function  $f$  to the value which is out side of a range  $[0, s-1]$  cannot return it to the original interval range.  $\square$

*Remark 1.* In case of multiple final states  $q \in Q_{\text{acc}}$ , where  $|Q_{\text{acc}}| = \tau$  the  $k$ -subsequence reachability problem can be reduced to  $\tau$  vector reachability problems for the same matrix semigroup but different target vectors following the above construction.

### 3 Linear recurrence automata without nested loops

In this subsection we show that reachability for an automata a single loop structure over different linear recurrences corresponds to reachability in a single linear recurrence of a higher order (depth) preserving the number of non-zero coefficients. At the same time we show that the subsequence reachability problem for a class of automata with a state structure consisting of multiple cycles connected in sequence is undecidable.

It is well known that the characteristic polynomial of the companion matrix  $C(a)$  (see (4)) of a linear recurrence  $a = (a_1, \dots, a_k)$  is given by  $\Phi_{C(a)}(x) = x^k - a_1x^{k-1} - a_2x^{k-2} - \dots - a_k$ . Conversely, for a polynomial  $f(x) = x^k - a_1x^{k-1} - a_2x^{k-2} - \dots - a_k$ , we say that  $a = (a_1, \dots, a_k)$  is the linear recurrence defined by  $f(x)$ .

**Lemma 1.** *Let  $A$  be a  $k \times k$  matrix and  $\mathbf{x}$  and  $\mathbf{y}$  be  $k$ -dimensional vectors. Let  $(w_t)_{t \in \mathbb{N}}$  be the sequence defined as  $w_t = \mathbf{x}^\top \cdot A^t \cdot \mathbf{y}$ . Then  $(w_t)_{t \in \mathbb{N}}$  satisfies the linear recurrence defined by the characteristic polynomial  $\Phi_A(x)$ .*

*Proof.* Let  $\Phi_A(x) = x^k - a_1x^{k-1} - a_2x^{k-2} - \dots - a_k$ . By the Cayley-Hamilton theorem, we have  $\Phi_A(A) = O$ , hence  $A^t \cdot \Phi_A(A) = A^{t+k} - a_1A^{t+k-1} - a_2A^{t+k-2} - \dots - a_kA^t = O$  and thus  $w_{t+k} - a_1w_{t+k-1} - a_2w_{t+k-2} - \dots - a_kw_t = \mathbf{x}^\top \cdot (A^{t+k} - a_1A^{t+k-1} - a_2A^{t+k-2} - \dots - a_kA^t) \cdot \mathbf{y} = 0$ .  $\square$

**Proposition 1.** *Let  $a^{(1)}, \dots, a^{(m)}$  be  $m$  linear recurrences of depth  $k$ , and  $(w_t)_{t \in \mathbb{N}}$  be a sequence generated by applying  $a^{(1)}, \dots, a^{(m)}$  successively and cyclically to the initial sequence  $(w_{k-1}, \dots, w_0)$ . Then  $(w_t)_{t \in \mathbb{N}}$  satisfies the linear recurrence of depth  $k \cdot m$  defined by the characteristic polynomial  $\Phi_{C(a^{(m)}) \dots C(a^{(1)})}(x^m)$ .*

*Proof.* For each  $r = 0, 1, \dots, m-1$ , by using Lemma 1, we show that the subsequence  $(w_t \mid t \equiv r \pmod{m})_{t \in \mathbb{N}} = (w_{mt+r})_{t \in \mathbb{N}}$  satisfies the linear recurrence defined by  $\Phi_{C(a^{(m)}) \dots C(a^{(1)})}(x)$  (we write  $\Psi(x)$  for short). Suppose  $0 \leq r < m$

$$\begin{aligned} \text{and then } w_{mt+r} &= (0 \cdots 0 \ 1) \begin{pmatrix} w_{mt+r+k-1} \\ w_{mt+r+k-2} \\ \vdots \\ w_{mt+r} \end{pmatrix} = \\ &= (0 \cdots 0 \ 1) C(a^{(r)}) \cdots C(a^{(1)}) \cdot (C(a^{(m)}) \cdots C(a^{(1)}))^t \cdot \begin{pmatrix} w_{k-1} \\ w_{k-2} \\ \vdots \\ w_0 \end{pmatrix}. \end{aligned}$$

Hence, by Lemma 1, the subsequence  $(w_{mt+r})_{t \in \mathbb{N}}$  satisfies the linear recurrence defined by  $\Psi(x)$ . Suppose  $\Psi(x) = x^k - a_1x^{k-1} - a_2x^{k-2} - \dots - a_k$ . Then we have

$$\begin{aligned} w_{mt+r} &= a_1w_{m(t-1)+r} + a_2w_{m(t-2)+r} + \dots + a_kw_{m(t-k)+r} \\ &= a_1w_{(mt+r)-m} + a_2w_{(mt+r)-2m} + \dots + a_kw_{(mt+r)-km} \end{aligned}$$

for each  $t \geq k$  and  $r = 0, 1, \dots, m-1$ . Thus  $(w_t)_{t \in \mathbb{N}}$  satisfies the recurrence relation

$$w_t = a_1w_{t-m} + a_2w_{t-2m} + \dots + a_kw_{t-km} \quad (8)$$

of depth  $k \cdot m$ , which is defined by  $\Psi(x^m) = x^{km} - a_1x^{km-m} - a_2x^{km-2m} - \dots - a_k = \Phi_{C(a^{(m)}) \dots C(a^{(1)})}(x^m)$ .  $\square$

*Example 1.* Let  $a^{(1)} = (1, 2)$  and  $a^{(2)} = (3, 4)$ . If we put  $w_0 = w_1 = 1$ , then the alternating applications of  $a^{(1)}$  and  $a^{(2)}$  to  $(w_1, w_0)$  generates the sequence

$$\begin{aligned} w_2 &= a^{(1)} \begin{pmatrix} w_1 \\ w_0 \end{pmatrix} = 1 \cdot 1 + 2 \cdot 1 = 3, \\ w_3 &= a^{(2)} \begin{pmatrix} w_2 \\ w_1 \end{pmatrix} = 3 \cdot 3 + 4 \cdot 1 = 13, \\ w_4 &= a^{(1)} \begin{pmatrix} w_3 \\ w_2 \end{pmatrix} = 1 \cdot 13 + 2 \cdot 3 = 19 \end{aligned}$$

Continuing this process, we have

$$(w_t)_{t \in \mathbb{N}} = (1, 1, 3, 13, 19, 109, 147, 877, 1171, 7021, \dots). \quad (9)$$

The companion matrices are

$$C(a^{(1)}) = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}, \quad C(a^{(2)}) = \begin{pmatrix} 3 & 4 \\ 1 & 0 \end{pmatrix}, \quad C(a^{(2)})C(a^{(1)}) = \begin{pmatrix} 7 & 6 \\ 1 & 2 \end{pmatrix}.$$

Thus we have  $\Phi_{C(a^{(2)})C(a^{(1)})}(x^2) = x^4 - 9x^2 + 8$ . In fact, the sequence (9) satisfies the linear recurrence relation  $w_t = 9w_{t-2} - 8w_{t-4}$ .

*Remark 2.* The depth of linear recurrence relation obtained in Proposition 1 is not minimal (in respect to the depth) in general. Indeed, the sequence (9) satisfies the relation  $w_t = -w_{t-1} + 8w_{t-2} + 8w_{t-3}$  of depth three. Alternative construction of a single linear recurrence has been considered in [18]. However, the linear recurrence obtained in Proposition 1 does not depend on the initial sequence  $w_{k-1}, \dots, w_0$  and has at most  $k$  non-zero coefficients.

**Proposition 2.** *1-Subsequence Reachability Problem for an automaton  $\mathcal{A} \in \mathcal{C}$  with a single loop constructed from  $m$  linear recurrences  $f_1, f_2, \dots, f_m$  of order  $k$  can be reduced to the reachability problem for a linear recurrence sequence of order  $k \cdot m$  with  $k$  non-zero coefficients.*

*Proof.* Let an automaton  $\mathcal{A} \in \mathcal{C}$  have a single loop with  $m$  states and  $m - 1$  edges labeled by different linear recurrences. By applying Proposition 1 the edges of the loop can be labeled with a single type linear recurrence of a larger order and then the length of this loop can be reduced to one edge. The right-hand side of the recurrence relation (8) has at most  $k$  non-zero coefficients, while its depth is  $k \cdot m$ . So, at most  $k$  values out of  $k \cdot m$  previous values will be used in each linear recurrence to generate the next value  $w_t$ .  $\square$

*Remark 3.* The reduction in the other direction is not always possible. For example the linear recurrence of the order  $k : u_n = u_{n-k}$  can not be decomposed to a finite sequence of linear recurrences of any smaller order than  $k$ .

**Proposition 3.** *A linear recurrence automata (without nested loops) can generate zero values in the positions corresponding to a non-semilinear set.*

*Proof.* Let us consider the linear recurrence automaton with five states  $s_1, \dots, s_5$ , initial state  $s_1$  and the following edges with correspondent linear recurrences:  
 $(s_1, s_2): u_n = u_{n-1} + u_{n-2}$        $(s_3, s_4): u_n = 2u_{n-2}$   
 $(s_2, s_1): u_n = u_{n-2}$        $(s_4, s_3): u_n = u_{n-2}$   
 $(s_1, s_3): u_n = u_{n-2}$        $(s_3, s_5): u_n = u_{n-1} - u_{n-2}$

The automaton can reach the state  $s_5$  only with the value  $u_n = 2^p - q$  starting from an initial value  $u_0 = 1, u_2 = 1$ , where  $p, q \in \mathbb{N}$  and in general the sequence is of the form  $1, 1, 2, 1, 3, 1, 4, 1, \dots, q, 1, q, 2^1, q, 2^2, \dots, q, 2^p, 2^p - q$ . The loop between  $s_1$  and  $s_2$  computes the value  $q$  reaching the subsequence  $1, q$  and the loop with states  $s_3$  and  $s_4$  compute the value  $2^p$  reaching the subsequence  $q, 2^p$  and the final transition to  $s_5$  compute the difference between the last two values.  $\square$

Now we show that 1-subsequence reachability problem for a class of automata with a linear structure of connected cycles in general is undecidable.

**Proposition 4.** *Let  $\mathcal{L}_1^y, \dots, \mathcal{L}_z^y$  be cyclic walks of a given length  $y \in \mathbb{N}$  based on a set of linear recurrences  $\mathcal{L}_1, \dots, \mathcal{L}_z$  that are sequentially connected by a set of linear recurrences  $P_1, \dots, P_{z-1}$  in a LRA  $\mathcal{A}$  to form walks for any  $i_j \in \mathbb{N}$ :*

$$\mathcal{L}_1^{y \cdot i_1} P_1 \dots P_{j_1} \mathcal{L}_2^{y \cdot i_2} P_{j_1+1} \dots P_{j_2} \mathcal{L}_3^{y \cdot i_3} P_{j_2+1} \dots P_{j_3} \dots P_{j_{z-2}} \dots P_{j_{z-1}} \mathcal{L}_z^{y \cdot i_z}.$$

*The 1-subsequence reachability problem for  $\mathcal{A}$  is undecidable.*

*Proof.* It is known that the matrix equation (aka Knapsack problem in  $\mathbb{Z}^{n \times n}$ )

$$M_1^{x_1} \dots M_r^{x_r} = O \tag{10}$$

is undecidable for sufficiently large  $r, n \in \mathbb{N}$ , where  $M_i \in \mathbb{Z}^{n \times n}$  and  $O$  is  $n \times n$  zero matrix [1]. We will construct Linear Recurrence Automata to encode the sequential multiplications of matrices in matrix equation which is undecidable in general <sup>6</sup>. By applying a technique from Theorem 1 we can convert each matrix  $M_i$  into a loop of  $n^2$  linear recurrences  $f_{i_1}, \dots, f_{i_{n^2}}$  of the order  $2n^2 - 1$  with  $n^2$

<sup>6</sup> A subclass of matrix equations has been shown to be decidable in [3].

states  $s_0, s_1 \dots s_{i_n^2-1}$  computing the values of a matrix product. Following the order of matrix equation (10) two loops corresponding to multiplication by  $M_i$  followed by  $M_j$  will be connected sequentially by an edge labeled with  $f_{j_1}$  from a state  $s_{i_1}$  to  $s_{j_2}$ . Finally by applying Proposition 1 we can replace a sequence of linear recurrences  $f_{i_1}, \dots, f_{i_{n^2}}$  by a loop with a single linear recurrence  $\mathcal{L}_i$  of the order  $n^2(2n^2 - 1)$ , but a single edge connecting  $i$ -th and  $j$ -th loops need to be replaced by recalculating next  $n^2(2n^2 - 1)$  values by a sequence of linear recurrences  $f_{j_1}, \dots, f_{j_{n^2(2n^2-1)}}$ . Also the initial sequence of  $n^2(2n^2 - 1)$  values need to be precomputed to start the automata. So from this directly follows that the  $n^2$ -subsequence reachability problem of reaching a sequence with  $n^2$  zero values for the constructed automata with linear recurrences of the order  $n^2(2n^2 - 1)$  and a constant  $y = n^2$  is undecidable.  $\square$

#### 4 Linear recurrence automata of the low order

**Theorem 3.** *1-Subsequence Reachability Problem for a Linear Recurrence Automata  $\mathcal{A} \in LRA_{\mathbb{Q}(i)}(s, m, 1)$ , i.e. with linear recurrences of the order one is decidable in polynomial time, where  $\mathbb{Q}(i)$  is Gaussian rational numbers.*

*Proof.* Let us consider an automaton with  $s$  states labelled with numbers from 1 to  $s$  and transition labelled by linear recurrences of depth 1, which corresponds to a multiplication by complex numbers, e.g. with the update in the form  $x_t = c \cdot x_{t-1}$ ,  $c \in \mathbb{C}$ . The configuration of an automata is a pair of a current state  $q$  and a value  $x_{t-1}$  can be represented as a vector of dimension  $s + 1$ :

$$(0, \dots, 0, 1, 0, \dots, 0, x_{t-1})^T$$

$\underbrace{\hspace{1.5cm}}_{q-1} \quad \underbrace{\hspace{1.5cm}}_{s-q}$

Every transition in the automata with an update  $x_t = c \cdot x_{t-1}$  will be represented by a  $(s + 1) \times (s + 1)$  matrices, where top  $s \times s$  block is a permutation matrix to change a position of 1 representing the state transition and the lower block of a single value in the position  $s, s$  that contains  $c$ . Every matrix is a monomial matrix with elements over a commutative semigroup and the vector reachability for the monomial matrix semigroup is decidable in polynomial time [16]. Since Gaussian rational numbers  $\mathbb{Q}(i)$  can be represented by commutative  $2 \times 2$  matrices with rational elements this completes the proof.  $\square$

**Theorem 4.** *1-Subsequence Reachability Problem for a Linear Recurrence Automata  $\mathcal{A} \in LRA_{\mathbb{Z}}(s, m, 2)$ , i.e. with linear recurrences of the order two is PSPACE-hard.*

*Proof.* The reachability problem for an affine register machine over integers is PSPACE-hard [12]. Affine register machine over integers is defined as an automaton with a finite number of states and the transition labelled by an affine updates of the form  $x' = ax + b$ , where  $a, b \in \mathbb{Z}$ . The reachability problem for this machine is the questions whether the value "y" can appear in the register with affine updates at the final state of a machine starting from an initial state and

the initial register value  $x_0$ . The affine update  $ax+b$  could be represented by the matrix  $M = \begin{pmatrix} b & a \\ 1 & 0 \end{pmatrix}$  applying to the vector  $\begin{pmatrix} x \\ 1 \end{pmatrix}$  and resulting in  $\begin{pmatrix} ax+b \\ 1 \end{pmatrix}$ . Also this can be seen as a sequential application of two linear recurrences of the order two. The first one is  $\begin{pmatrix} b & a \\ 1 & 0 \end{pmatrix} \begin{pmatrix} u_{n-1} \\ u_{n-2} \end{pmatrix} = \begin{pmatrix} u_n \\ u_{n-1} \end{pmatrix}$  and the second one  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} u_{n-1} \\ u_{n-2} \end{pmatrix} = \begin{pmatrix} u_n \\ u_{n-1} \end{pmatrix}$  and starting with initial values  $\begin{pmatrix} 1 \\ x_0 \end{pmatrix}$ . We can construct linear recurrence automaton from the register machine with affine updates by inheriting the state structure with the same initial and the final state and by replacing each transition  $ax+b$  by two transitions with sequential linear recurrences  $u_n = bu_{n-1} + au_{n-2}$  and  $u_n = u_{n-2}$ . So by applying in alternative way the first and the second linear recurrence we generate the sequence  $x_0, 1, a(x_0) + b, 1, a(a(x_0) + b) + b, 1, \dots$   $\square$

*Remark 4.* So the 1-subsequence reachability problem for a Linear Recurrence Automata is at least PSPACE-hard and upto our knowledge it is not known whether this problem with linear recurrences of the order (depth) two is decidable or not. Following Theorem 2 the 2-subsequence reachability could be reduced to the vector reachability for  $4 \times 4$  matrices which is in general undecidable [17,8].

**Theorem 5.** *2-Subsequence Reachability Problem for a Linear Recurrence Automata  $\mathcal{A} \in LRA_{\mathbb{Q}}(7, 11, 3)$ , i.e. with linear recurrences of the order three is undecidable.*

*Proof.* It was shown in [2] that the point-to-point reachability is undecidable for a set of five affine transformations of dimension two over rational numbers. In fact a class of affine transformations of dimension two with undecidable reachability can be restricted to the following form:

$$f(p, q) = \begin{cases} p' = ap + bq + c \\ q' = dq \end{cases}$$

which corresponds to undecidable vector reachability for matrix semigroups generated by  $3 \times 3$  matrices over rational numbers

$$M = \begin{pmatrix} a & b & c \\ 0 & d & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ applying to the vector } \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

with starting from a vector  $(0, 1, 1)^T$  and target vector  $(0, 1, 1)^T$ . Following construction of the Theorem 1 we can build an automaton that can generate the sequence  $p_0, q_0, 1, p_1, q_1, 1, p_2, q_2, 1, \dots, p_i, q_i, 1$ . Moreover 2-subsequence reachability problem and in particular reaching 2-subsequence 0, 1 at any position is undecidable since it can only appear when 0 is in the position  $i = 0(\text{mod}3)$  and 1 in the position  $i = 1(\text{mod}3)$  based on the matrix form. At the same time undecidability holds for reaching 1-subsequence 0 at the position  $i = 0(\text{mod}3)$

which will correspond to reaching 0 when automaton in the final state which is also an initial state of the constructed automaton. Note that the linear recurrences for one iteration of matrix multiplications will be of order 3 and the last transformation to copy value 1 in the position  $i = 2(\bmod 3)$  is the same for any matrix  $M$ . So we have three sequential transformations

$$x_j = ax_{j-3} + bx_{j-2} + cx_{j-1} \rightarrow x_j = dx_{j-3} \rightarrow x_j = x_{j-3}.$$

In this case only  $m+2$  states and  $2m+1$  recurrences of order 3 are needed, where  $m$  is the number of affine transformations and therefore we have undecidability for automata  $\mathcal{A} \in LRA_{\mathbb{Q}}(7, 11, 3)$ .  $\square$

**Theorem 6.** *The reachability for stateless polynomial recurrent system in dimension two is undecidable.*

*Proof.* It is known that the reachability problem is undecidable for stateless affine automata in dimension three and for two dimensional affine automata with states where updates are of the form:  $x' = ax + b$  and  $y' = cy + d$  [14]. Two dimensional affine transformations can be encoded by one dimensional recurrence automata with states and the sequence  $x_1, y_1, 1, x_2, y_2, 1, \dots$  by a sequence of recurrences

$$u_n = a \cdot u_{n-3} + b \cdot u_{n-1} \rightarrow u_n = c \cdot u_{n-3} + d \cdot u_{n-2} \rightarrow u_n = u_{j-n}.$$

In the second dimension it is possible to encode transitions between states with a polynomial updates where state  $i$  can be encoded as  $10^i$  and then transition from state  $i$  to state  $j$  is encoded by a function  $v_n = \frac{v_{n-1}^s}{10^{((s-i-j))}}$ .  $\square$

## 5 Multidimensional linear recurrences

Another extension of the automata model that can be considered is a case of multidimensional linear recurrences which can be seen as a generalisation of standard linear recurrences. For example, in the biological context the model can be used to describe the populations of a group of highly interconnected species or organisms (e.g. symbiotic bacteria). Let us define a sequence  $X_0, X_1, X_2, \dots$  of  $r \times s$ -matrices over  $\mathbb{F}$  is called *linearly recursive of depth  $k$*  or *recursion* or *recurrence* if there are  $r \times r$  matrices  $C_1, \dots, C_k$  so that  $X_n = C_1 X_{n-1} + C_2 X_{n-2} + \dots + C_k X_{n-k}$  for  $n > k$ . Let us then consider a number  $i \in \{1, \dots, m\}$  of linear recurrences

$$X_n = C_1^{(i)} X_{n-1} + C_2^{(i)} X_{n-2} + \dots + C_k^{(i)} X_{n-k}. \quad (11)$$

The reachability problems for automata with multidimensional linear recurrences of the order two are undecidable for dimension four when  $C_i$  are diagonal by simulation of reachability in independent affine updates [14] alternating affine updates with copying constant value like in Theorem 4. Also for dimension three and for more general case of  $C_j^i$  values the stateless automata has undecidable reachability problem based on [2,9]. However in case of limiting the matrix structure or parameters the undecidability results would not hold and more general tools for analysis of this model may be required.

**Proposition 5.** *A sequence  $X_0, X_1, X_2, \dots$  of  $r \times s$  matrices is linearly recursive if and only if there are matrices  $A$  ( $r \times d$ ),  $M$  ( $d \times d$ ), and  $B$  ( $d \times s$ ) so that for  $n \geq 0$*

$$X_n = AM^n B. \quad (12)$$

*Proof.* Assume first (11) with only one recurrence  $C_j = C_j^{(i)}$ , let  $d = kr$  and create a  $kr \times kr$  companion matrix in block form

$$M = \begin{pmatrix} C_1 & C_2 & \dots & C_{k-1} & C_k \\ I & 0 & \dots & 0 & 0 \\ 0 & I & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & I & 0 \end{pmatrix}.$$

It follows immediately that

$$M \begin{pmatrix} X_{n-1} \\ X_{n-2} \\ \vdots \\ X_{n-d} \end{pmatrix} = \begin{pmatrix} X_n \\ X_{n-1} \\ \vdots \\ X_{n-d+1} \end{pmatrix},$$

and by induction

$$M^n \begin{pmatrix} X_{d-1} \\ X_{d-2} \\ \vdots \\ X_0 \end{pmatrix} = \begin{pmatrix} X_{d+n} \\ X_{d+n-1} \\ \vdots \\ X_n \end{pmatrix}$$

Letting  $A = (0 \ 0 \dots I)$  ( $r \times kr$  matrix) and  $B = \begin{pmatrix} X_{d-1} \\ X_{d-2} \\ \vdots \\ X_0 \end{pmatrix}$  ( $kr \times s$  matrix) we

see that

$$X_n = AM^n B.$$

Assume then (12) holds for some  $d \times d$ -matrix  $M$ . By Cayley-Hamilton theorem,

$$M^d - c_1 M^{d-1} - c_2 M^{d-2} - \dots - c_{d-1} M - c_d I = 0, \quad (13)$$

where  $p(x) = x^d - c_1 x^{d-1} - c_2 x^{d-2} - \dots - c_{d-1} x - c_d$  is the characteristic polynomial of matrix  $M$ .

Multiplying (13) by  $M^{n-k}$  we get

$$M^n = c_1 M^{n-1} + c_2 M^{n-2} + \dots + c_d M^{n-d}, \quad (14)$$

and left and right multiplication of (14) by  $A$  and  $B$  respectively gives

$$X_n = c_1 X_{n-1} + c_2 X_{n-2} + \dots + c_d X_{n-d}.$$

Claim follows now by choosing  $C_i = c_i I$ , where  $I$  is  $r \times r$  identity matrix.  $\square$

**Definition 4.** Sequence  $X_0, X_1, X_2, \dots$  matrices is said to be nondeterministically recurrent or recurrent through choices  $i_0, i_1, i_2, \dots$  if for  $n > k$   $X_n$  is obtained by recurrence (11) by selecting the indices  $i_k$  nondeterministically in set  $\{1, \dots, m\}$ . If  $m = 1$ , the sequence is said to be deterministically recurrent.

**Definition 5.** Let  $w = a_1 a_2 \dots a_l$  be a word over alphabet  $\Sigma = \{1, \dots, m\}$  and  $M_1, \dots, M_m$  a set of matrices indexed by the elements of  $\Sigma$ . Then we define

$$M_w = M_{a_1} M_{a_2} \dots M_{a_l},$$

which is just another way of expressing the fact that  $w \rightarrow M_w$  is a morphism from  $\Sigma^*$  to the semigroup generated by matrices  $M_a, a \in \Sigma$ .

**Proposition 6.** Sequence  $X_n$  of  $r \times s$  matrices is linearly recurrent through choices  $i_0, i_1, i_2, \dots$  if and only if

$$X_n = A M_{w^R} B, \quad (15)$$

where  $w = i_0 i_1 \dots i_l \in \{1, \dots, m\}^*$  and  $w^R = i_l i_{l-1} \dots i_1 i_0$  stands for the mirror image of  $w$ ,  $A$  is a  $r \times d$ -matrix, each  $M_i$  is a  $d \times d$ -matrix, and  $B$  is a  $d \times s$ -matrix.

*Proof.* Assume first that there are  $m$  linear recurrences as in (11). By convention, we suppose that all those have their  $k$  initial values  $X_0^{(i)}, \dots, X_{k-1}^{(i)}$ . Matrix  $B$  in the claim is constructed as a block matrix  $B = (X_{k-1}^{i_0} \dots X_0^{i_0})^T$ , where  $i_0, i_1, \dots, i_{k-1}$  are selected from the set  $\{1, 2, \dots, m\}$ . Each matrix  $M_i$  and  $A$  are constructed as a block-formed companion matrix of  $i$ :th recurrence just like in Proposition 5 and the claim follows straightforwardly.

For the opposite direction, we need to show that each finite set of matrices  $\{M_1, \dots, M_m\}$  gives raise to a set of linear recurrences via equation (15).

Multiple choices for recurrence means that we do not have only one sequence  $B \rightarrow MB \rightarrow M^2 B, \dots$  as in Proposition 5, but rather a tree with  $B$  as a root, and all  $M_i B$  as nodes with distance 1 from the root, all  $M_j M_i B$  as nodes with distance 2 from the root, all nodes  $M_k M_j M_i B$  with distance 3 from the root, etc., see Figure 1.

To generalize, we define  $D_i = \{M_w B \mid |w| = i\}$  as the set of all nodes with distance  $i$  from the root.

If  $D = D_1 \cup D_2 \cup \dots$  is finite, only finitely many matrices of form  $M_w B$  exist. Then for each  $w$  long enough and each  $i$ , we have  $M_i M_w B \in D$ , and hence the set of recursions is trivial.

We can therefore assume that set  $D$  is infinite. Since set  $D \subseteq \mathbb{F}^{d \times s}$  (a vector space of dimension  $d \cdot s$ ),  $D$  generates a subspace  $V \subseteq \mathbb{F}^{d \times s}$ , and hence there exists a basis  $\mathcal{B} \subset D$  of  $V$  of cardinality at most  $ds$ . The basis is by no means unique, but we choose a basis  $\mathcal{B}$  so that  $\max\{|w| \mid M_w B \in \mathcal{B}\}$  is minimal. Now, the set  $\{A M_w B \mid M_w B \in \mathcal{B}\}$  is considered as the set of initial values of recurrence, and for longer words, the recurrence is obtained in the following way:

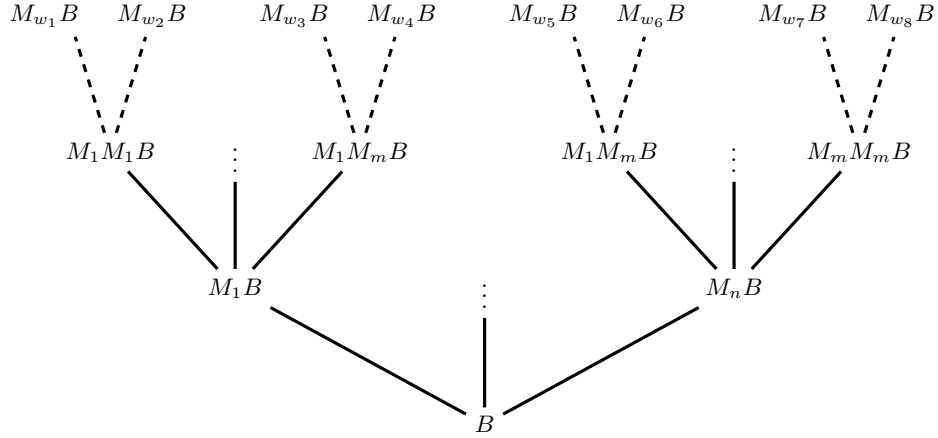
For each matrix  $M_i$  and each  $M_w B \in \mathcal{B}$ , matrix  $M_i M_w B$  belongs to the subspace  $V$ , and hence

$$M_i M_w B = \sum_{M_v B \in \mathcal{B}} c_{M_v}^{(i)} M_v B,$$

where coefficient  $c_{M_v} \in \mathbb{F}$ , and hence value  $AM_iM_wB$  can be obtained by using initial values:

$$AM_iM_wB = \sum_{M_v B \in \mathcal{B}} c_{M_v}^{(i)} AM_v B.$$

□



**Fig. 1.** A tree of possibilities with multiple choices for recurrences.

**Corollary 1.** *If  $X_n$  and  $Y_n$  are recurrent sequences (deterministically or non-deterministically), so are*

1.  $X_n \otimes Y_n$  (tensor product).
2.  $X_n + Y_n$ , if the dimensions of  $X_n$  and  $Y_n$  match.
3.  $CX_n D$ , if the dimensions of  $C$  and  $D$  are compatible.

The next statement follows from Propositions 5 and 6 since  $X_n = AM_{w_1^R}^n M_{w_0^R} B = AM_{w_1^R}^n B_1$ , where  $B_1 = M_{w_0^R} B$ .

**Corollary 2.** *Let  $X_n$  be a sequence obtained by nondeterministic linear recurrences in set  $\{1, 2, \dots, m\}$  appearing eventually in cyclic order:  $w = w_0 w^n$ , where  $w_0 \in \{1, 2, \dots, m\}$  is arbitrary. Then there is a deterministic linear recurrence generating sequence  $X_n$ .*

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